SEAT No.________

[72/A-25]

No. of printed pages: 02

SARDAR PATEL UNIVERSITY

SYBCA (Semester - IV)

Operating Systems (US04CBCA03)

Date: 16/6/201- Total Marks: 70

Time: 2.00 p.m. to 5.00 p.m.

Q.1 Pick up the correct alternative for each of the following questions: **[10]**

1. Patient Monitoring in hospital is an example of which type of OS?
   - A. Multi User B. Time Sharing
   - C. Real Time D. None
2. A Program in execution is known as _______.
   - A. Process B. Commands
   - C. Files D. None of these
3. In monolithic structure, the OS divided into number of ______.
   - A. Files B. Procedures
   - C. Functions D. Programs
4. Technique use to solve problem of external fragmentation is known as ___________.
   - A. Process B. Aging
   - C. Compaction D. None of these
5. A Lazy Swapper is also known as ________.
   - A. Pager B. Pages
   - C. Frames D. None of these
6. A_____________ process produces information that is consumed by a consumer process.
   - A. Consumer B. Computation
   - C. Producer D. None
7. Printer is an example of ________ resource.
   - A. Logical B. Virtual
   - C. Physical D. None
8. _______ option of vi editor is use to save your work.
   - A. cp B. rm
   - C. mv D. :w
9. _________ command is use to display list of all users currently login on a server.
   - A. grep B. search
   - C. fine D. None
10. Case statement ends with ______.
    - A. end case B. esac
    - C. end D. none of these

Q.2 Attempt any ten from following: **[20]**

1. Define Operating system.
2. Define Throughput.
3. What is Context Switching?
4. What is Belady's Anomaly?
5. Explain First-fit memory allocation techniques.
6. Explain Best-fit memory allocation techniques.

7. When Race conditions arise?
8. Explain any two necessary conditions of Critical section.
9. Why is LINUX an open source?
10. Explain man command in short.
11. Explain who command in brief.
12. Explain ls –l command.

Q.3	[A]	Explain process state and PCB in brief.	[05]
	[B]	Describe FCFS scheduling with advantages and disadvantages.	[05]
		OR	
Q.3	[A]	Explain SJF scheduling algorithm in brief.	[05]
	[B]	Explain Round-Robin scheduling algorithm with example.	[05]
Q.4	[A]	Explain Swapping of two processes with suitable example.	[05]
	[B]	Explain concept of Memory Compaction. What are the problems with this approach?	[05]
		OR	
Q.4	[A]	Explain FIFO page replacement algorithm with advantages and disadvantages.	[05]
	[B]	Explain Demand Paging.	[05]
Q.5	[A]	What is LINUX? Explain basic features of LINUX Operating System.	[05]
	[B]	Define Deadlock. Explain all necessary conditions for occurrence of deadlock.	[05]
		OR	
Q.5	[A]	Define Cooperative Process. Explain Producer-Consumer Problem.	[05]
	[B]	List out differences between cooperative and independent process. Also explain advantages of cooperative process.	[05]
Q.6		Explain grep command with at least five possible attributes and examples.	[10]
		OR	
Q.6		Explain all forms of IF.	[10]

***** X *****

\22